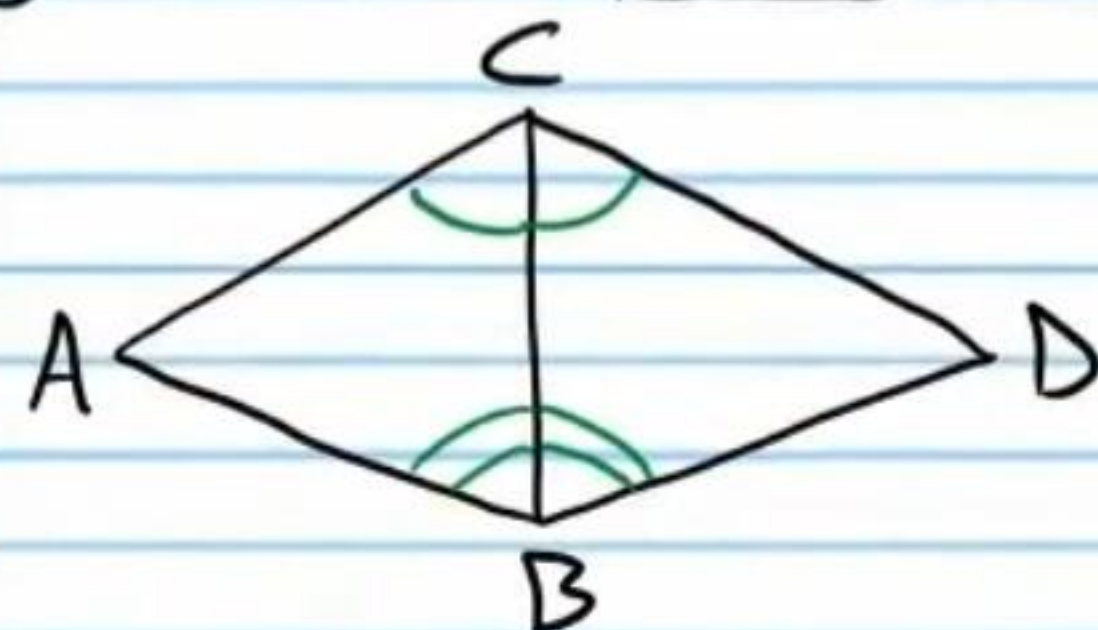


Triangle Congruence Postulates and Theorems (Homework)

In the problems below, determine if the triangles are congruent. If they are congruent, write a proper congruence statement and the triangle postulate or theorem that proves that they are congruent. You do not have to write a formal proof (i.e. a two-column proof).

① $\triangle ABC \cong$?

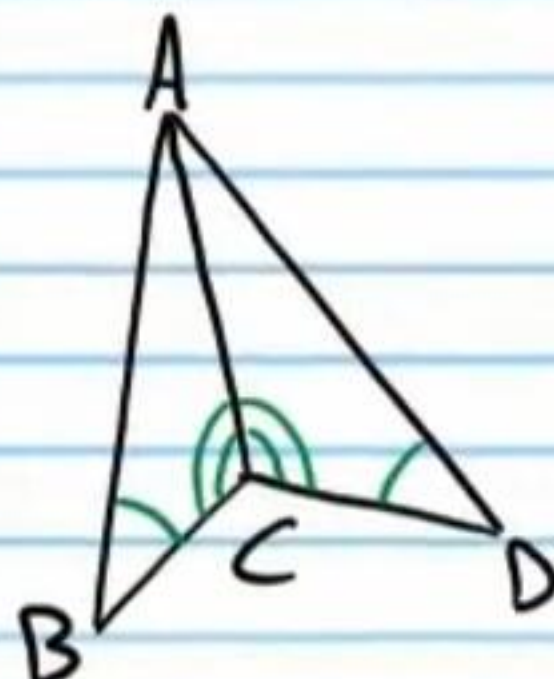


Answers

$\triangle ABC \cong \triangle DAC$

ASA

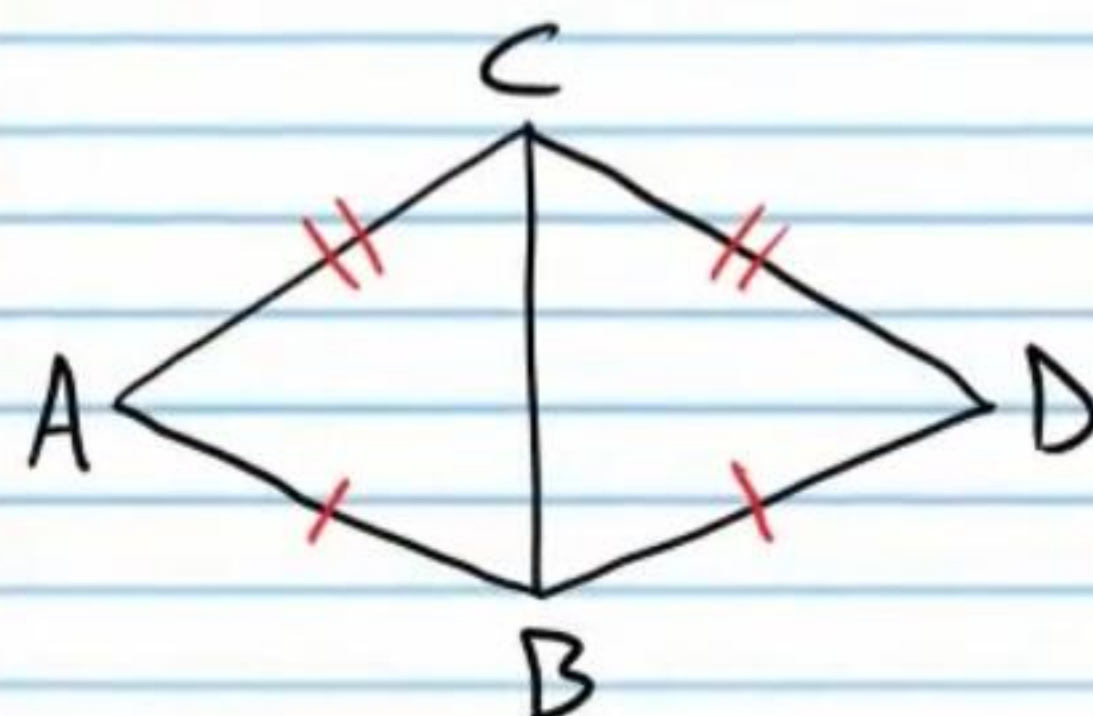
② $\triangle ABC \cong$?



$\triangle ABC \cong \triangle ADC$

AAS

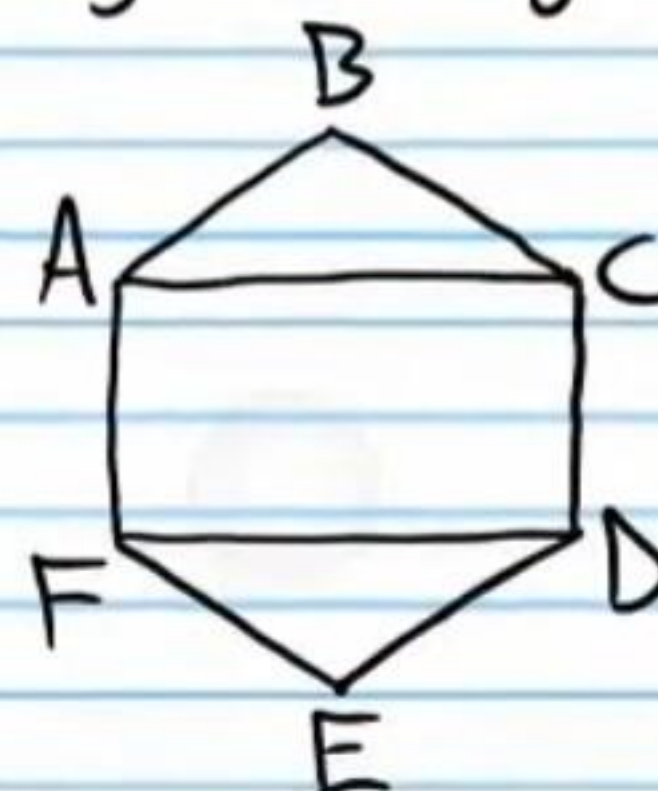
③ $\triangle ABC \cong$?



$\triangle ABC \cong \triangle DAC$

SSS

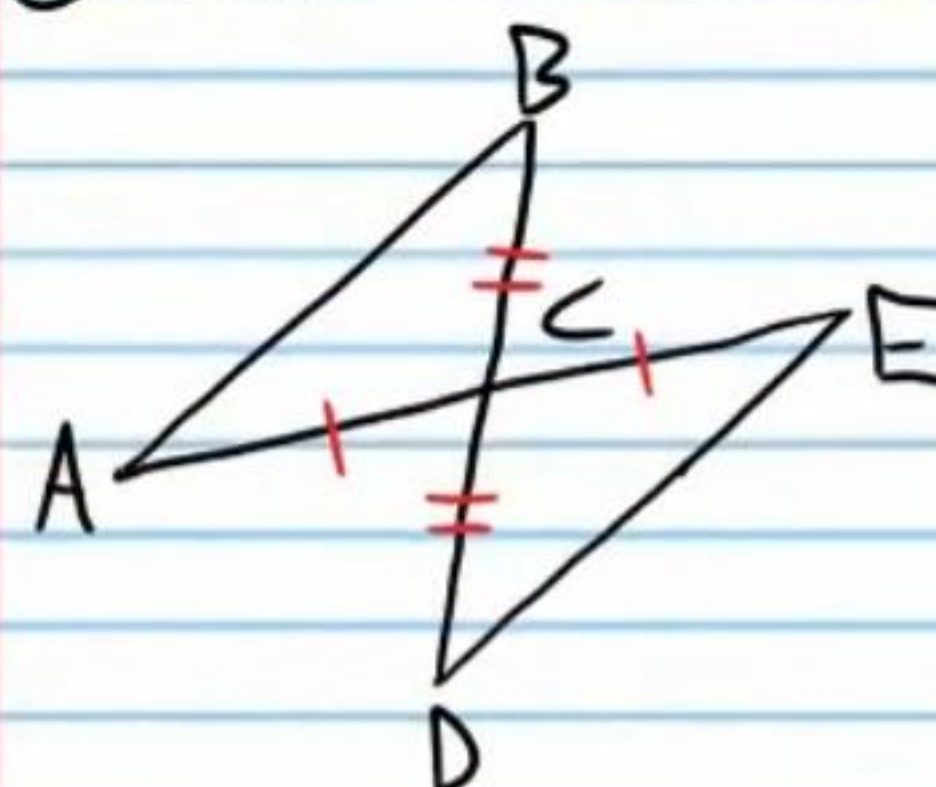
④ $\triangle ABC \cong$?
given that ABCDEF is a regular hexagon.



$\triangle ABC \cong \triangle FED$

SAS

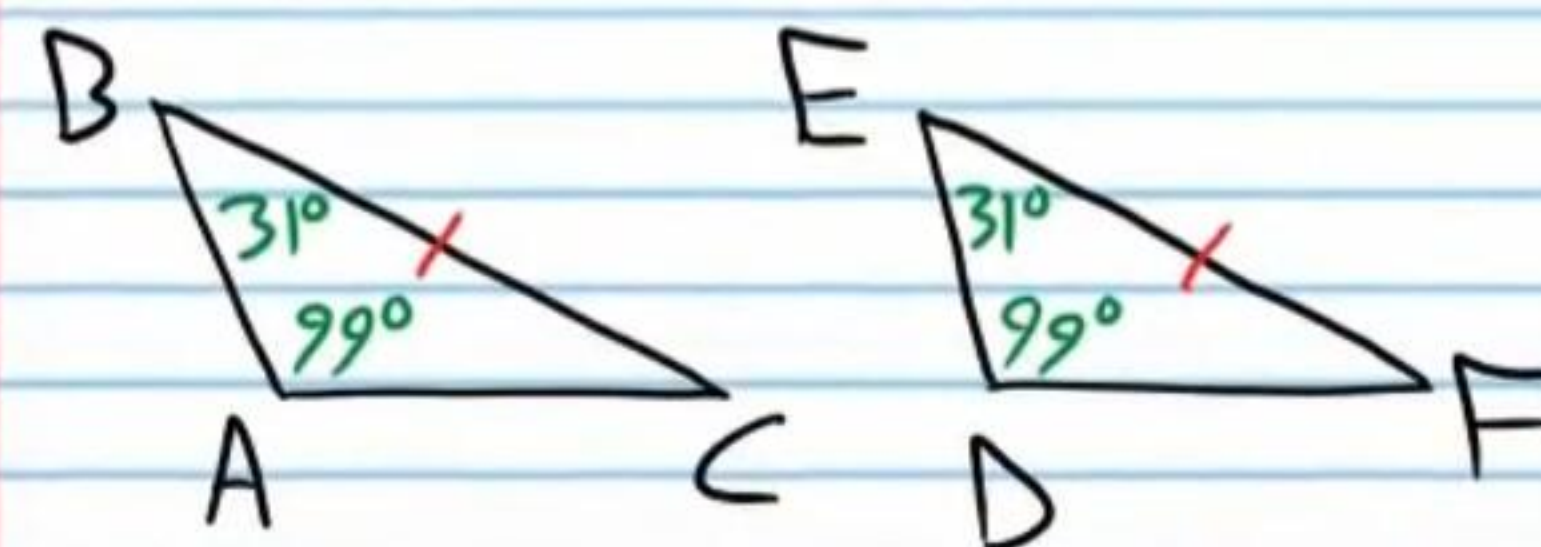
⑤ $\triangle ABC \cong$?



$\triangle ABC \cong \triangle EDC$

SAS

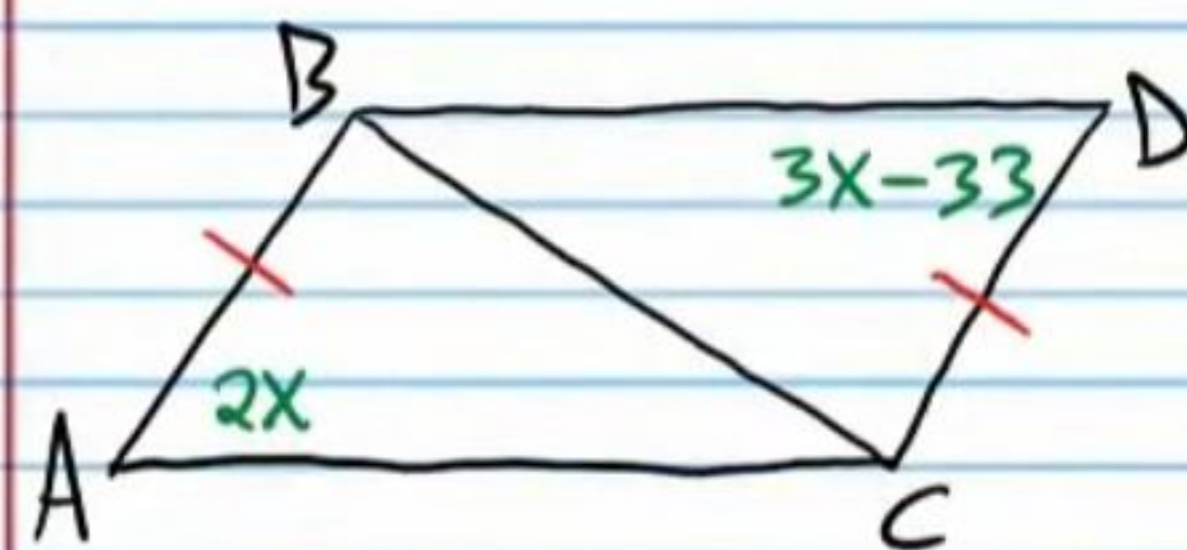
⑥ $\triangle ABC \cong$?



$\triangle ABC \cong \triangle DEF$

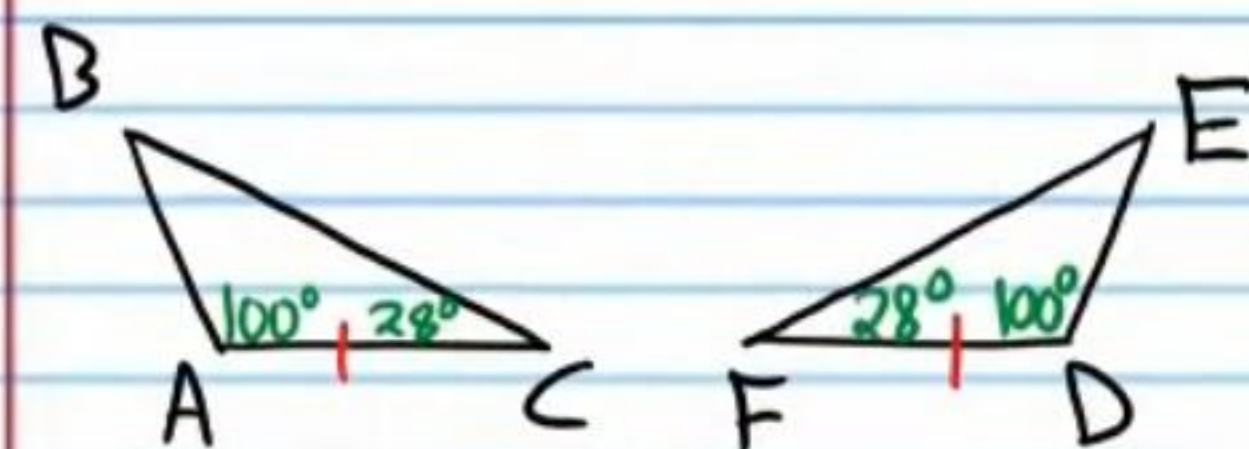
AAS

⑦ $\triangle ABC \cong$?



Not enough information

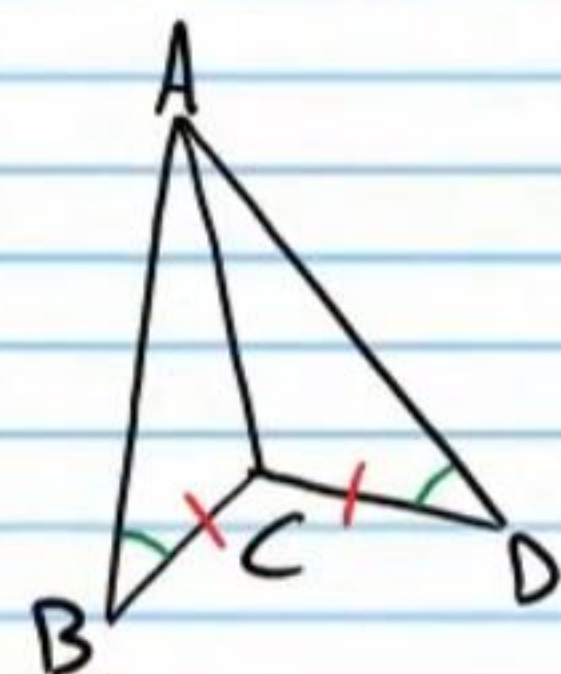
⑧ $\triangle ABC \cong$?



$\triangle ABC \cong \triangle DEF$

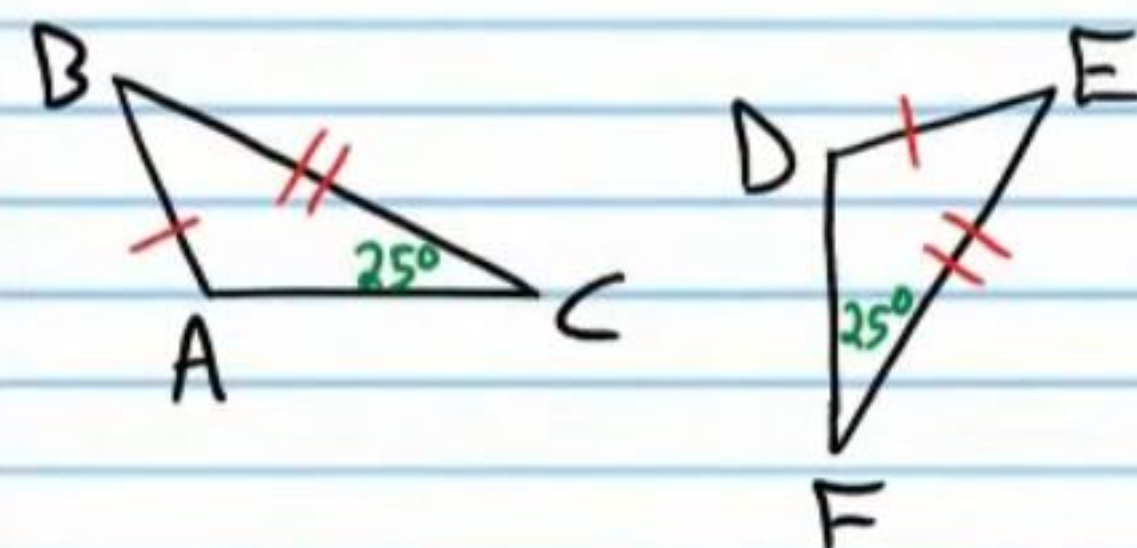
ASA

⑨ $\triangle ABC \cong$?



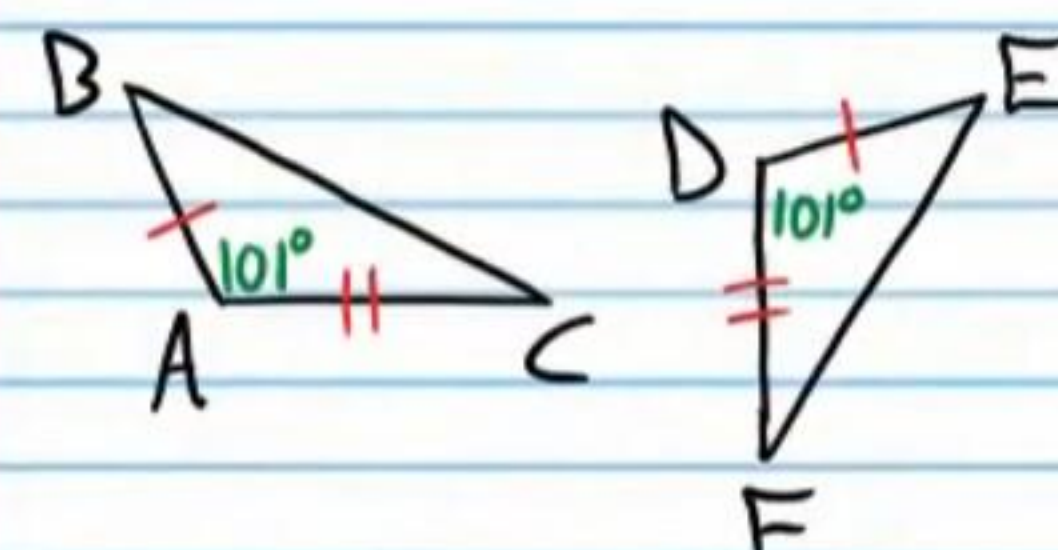
Not enough information.

⑩ $\triangle ABC \cong$?



Not enough information

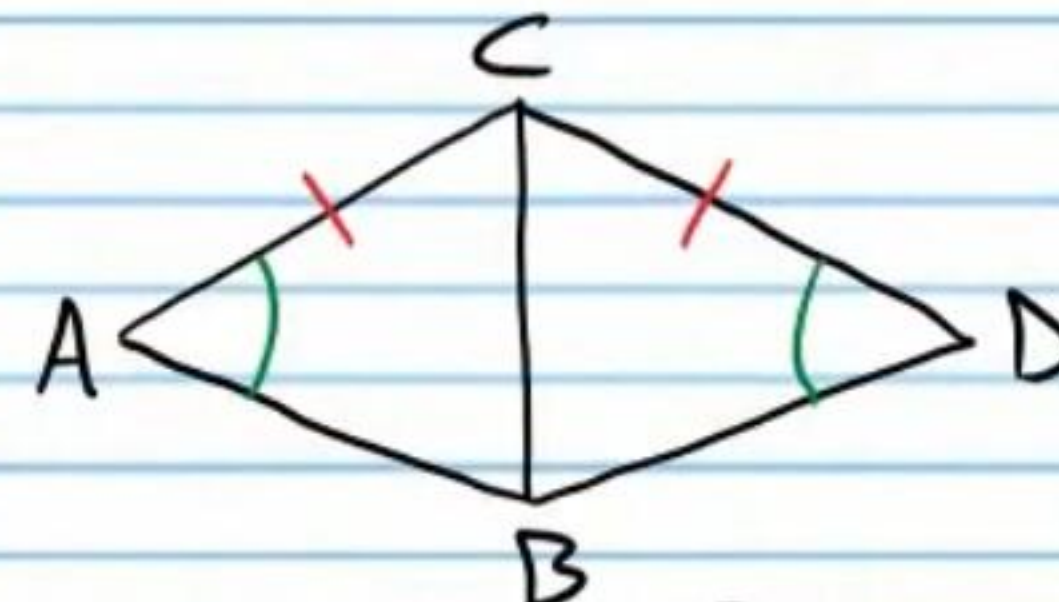
⑪ $\triangle ABC \cong$?



$\triangle ABC \cong \triangle DEF$

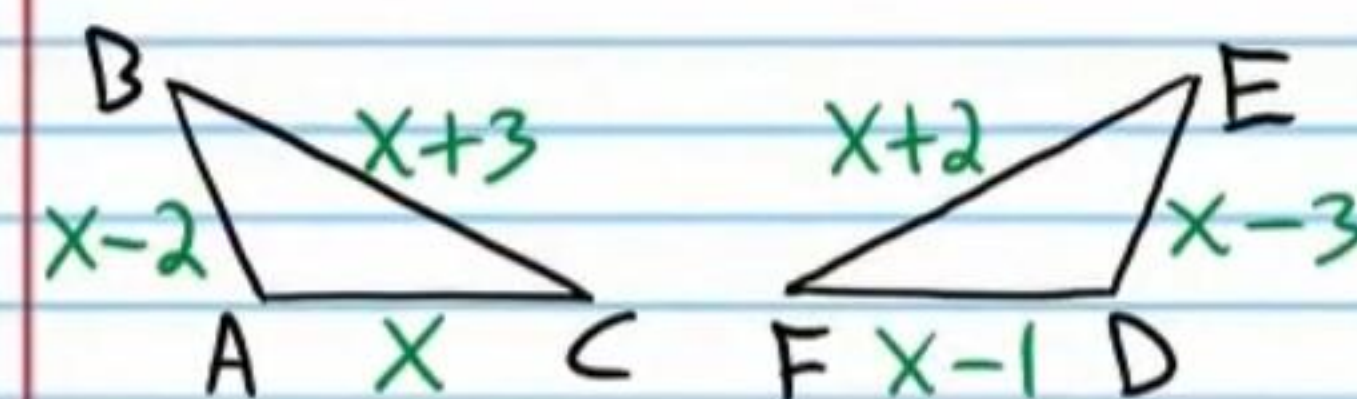
SAS

⑫ $\triangle ABC \cong$?



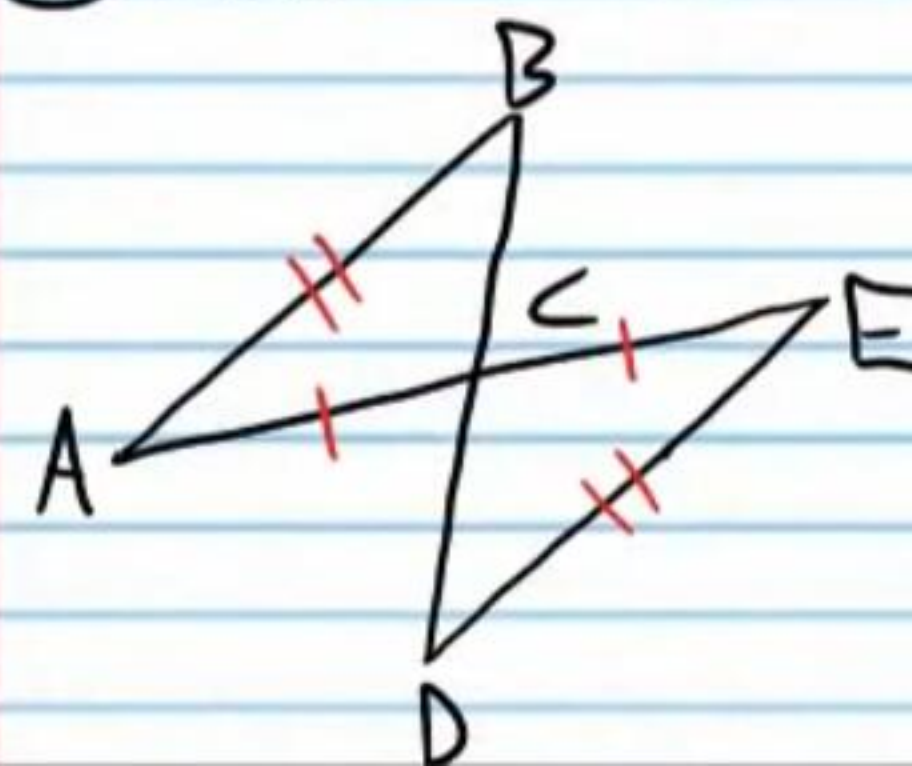
Not enough information

⑬ $\triangle ABC \cong$? if $x=8$.



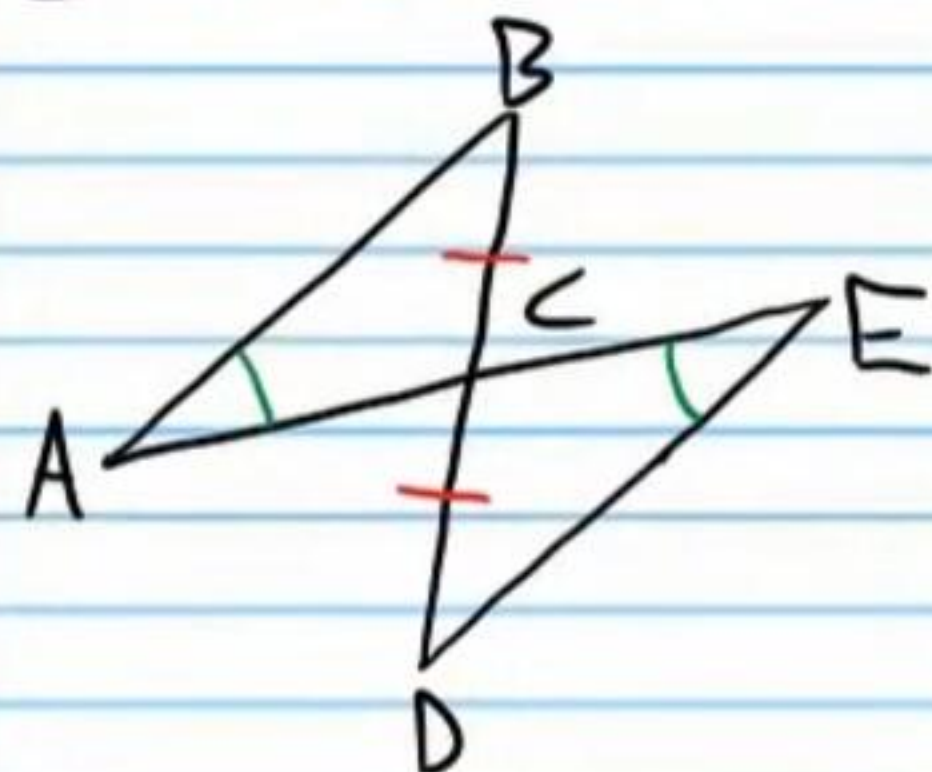
Not $\cong$

⑭ $\triangle ABC \cong$?



Not enough information

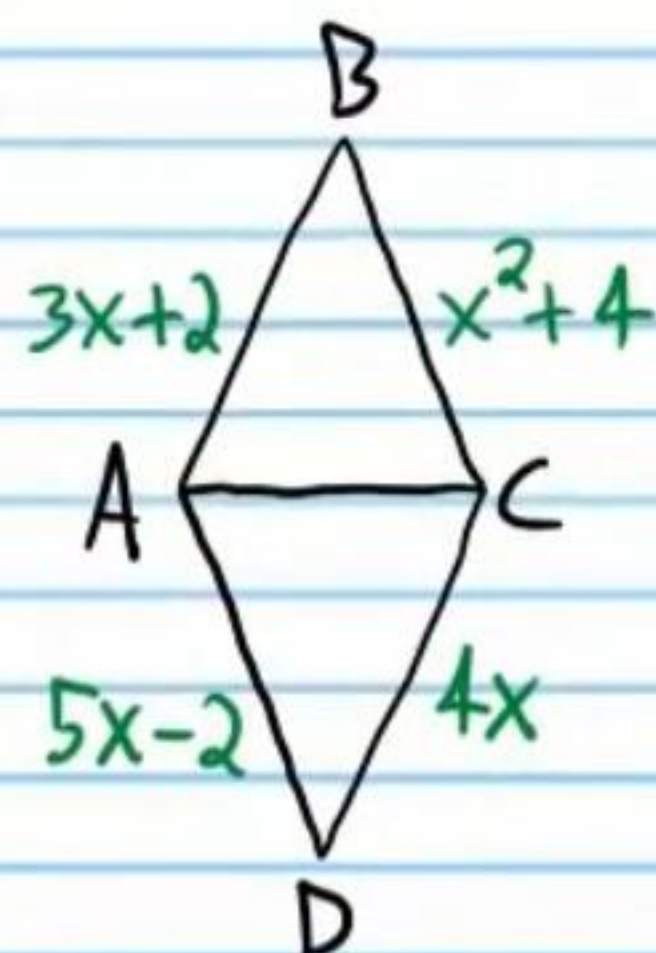
⑮ $\triangle ABC \cong$?



$$\triangle ABC \cong \triangle EDC$$

AAS

⑯ $\triangle ABC \cong$? if $x=2$.

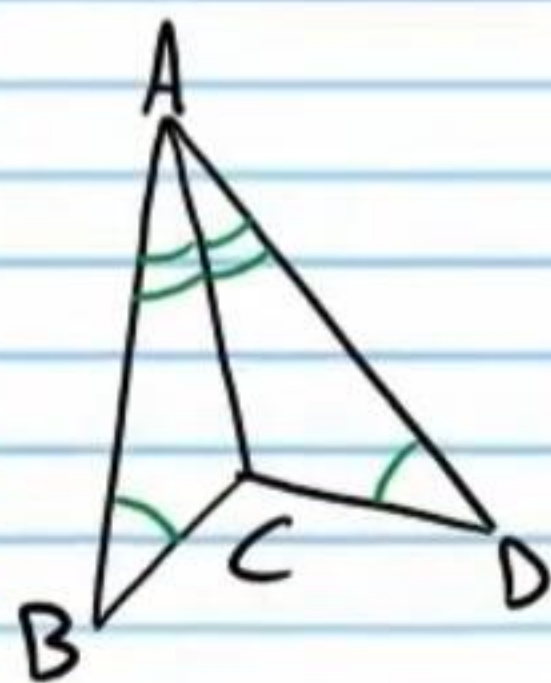


$$\triangle ABC \cong \triangle CDA$$

SSS

ADC is also
an answer.

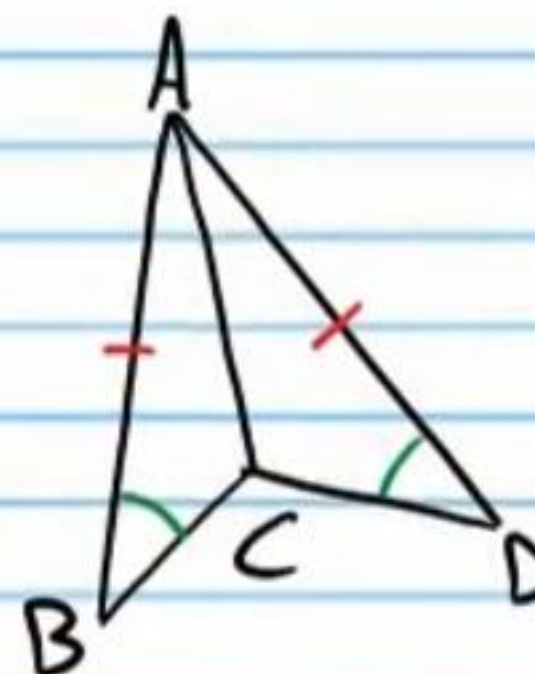
⑰ $\triangle ABC \cong$?



$$\triangle ABC \cong \triangle ADC$$

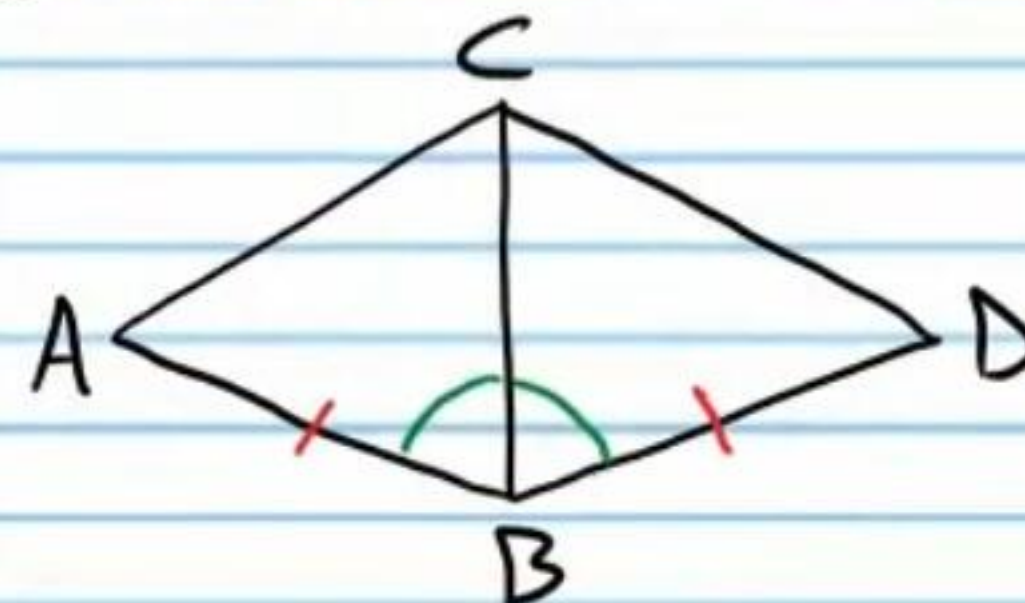
AAS

⑱ $\triangle ABC \cong$?



Not enough
information

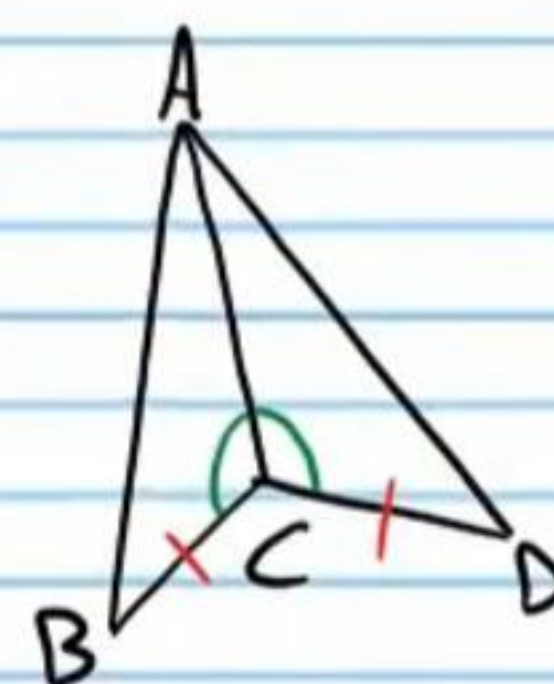
⑲ $\triangle ABC \cong$?



$$\triangle ABC \cong \triangle ADC$$

SAS

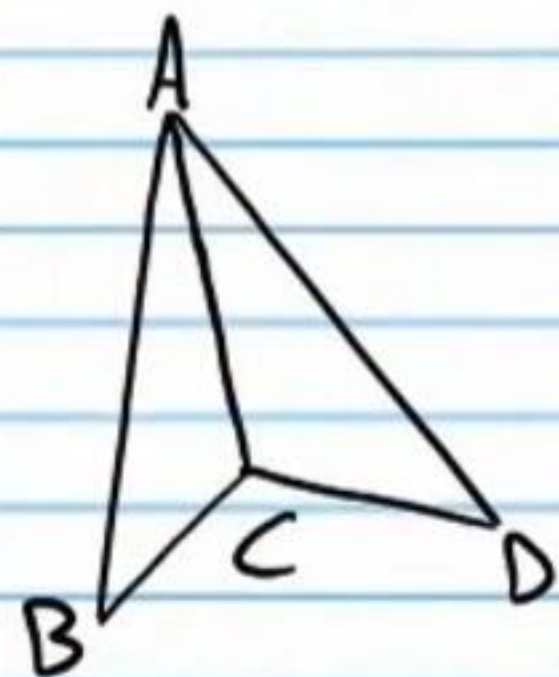
⑳ $\triangle ABC \cong$?



$$\triangle ABC \cong \triangle ADC$$

SAS

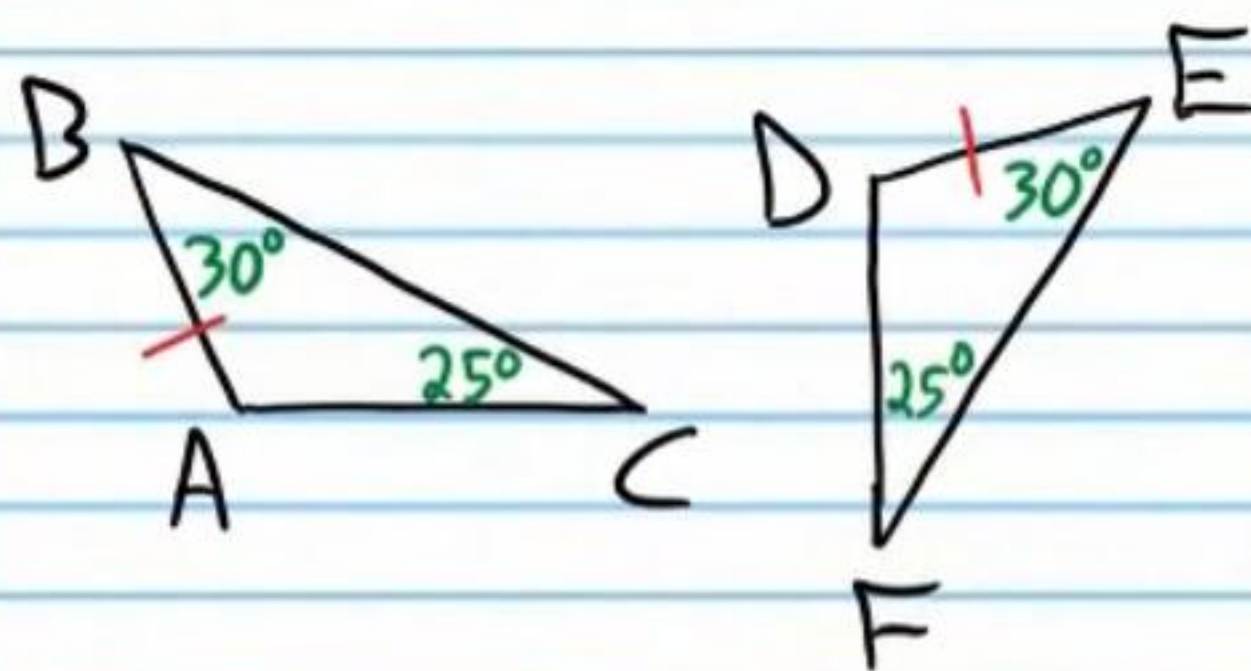
② $\triangle ABC \cong$?
given that $\overline{AC}$ bisects $\angle BAD$ and also $\angle BCD$.



$$\triangle ABC \cong \triangle ADC$$

ASA

② $\triangle ABC \cong$?



$$\triangle ABC \cong \triangle DEF$$

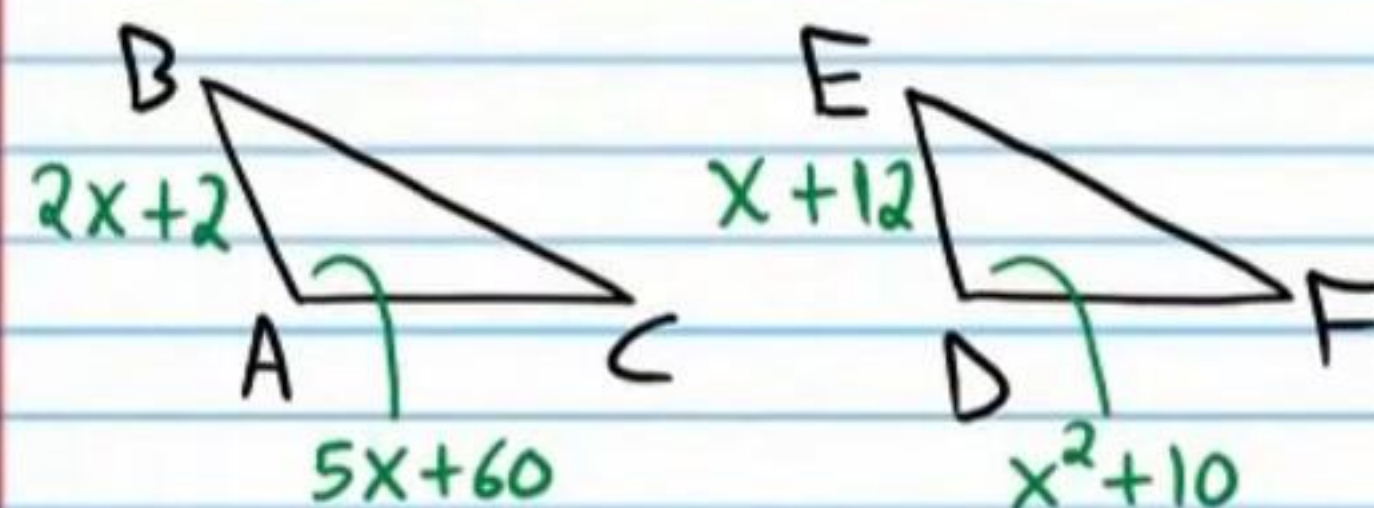
AAS

② $\triangle ABC \cong$?



Not enough information

② $\triangle ABC \cong$? if $x=10$ and $\overline{AC} \cong \overline{DF}$.

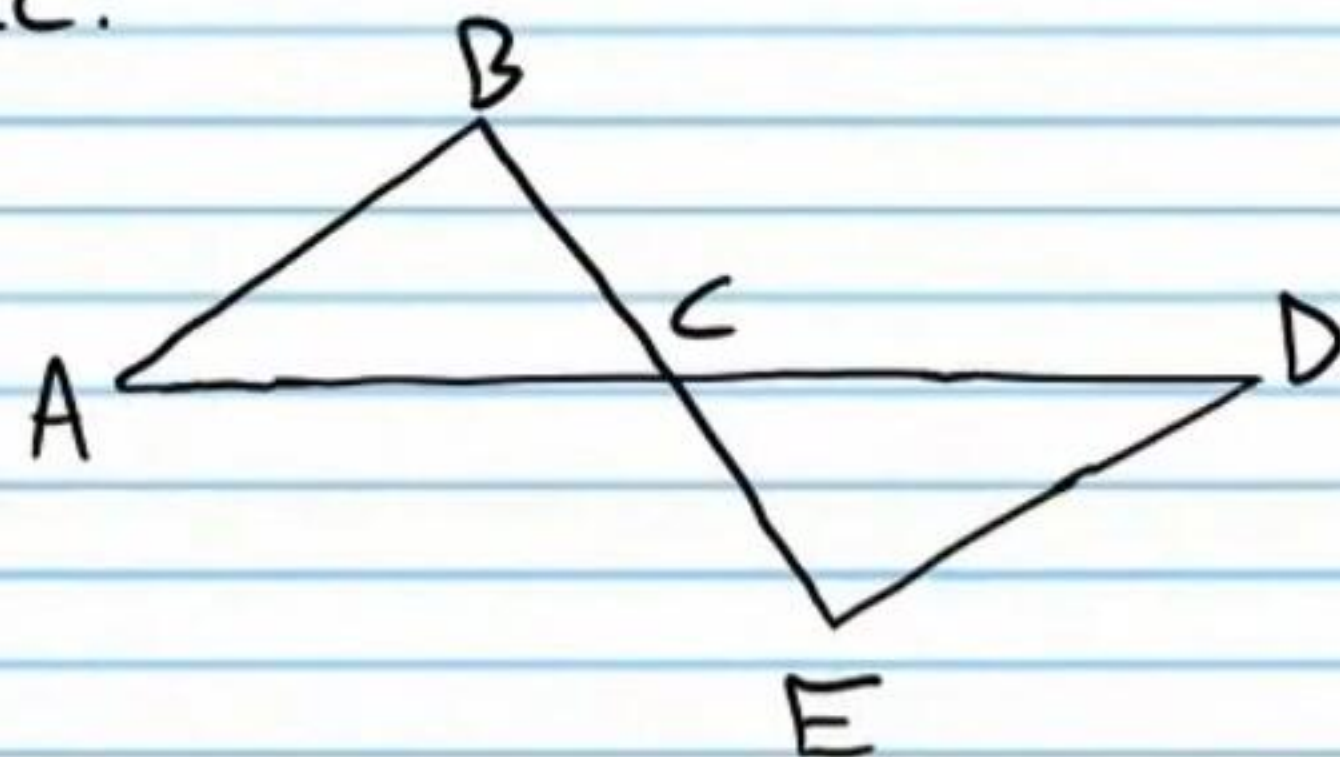


$$\triangle ABC \cong \triangle DEF$$

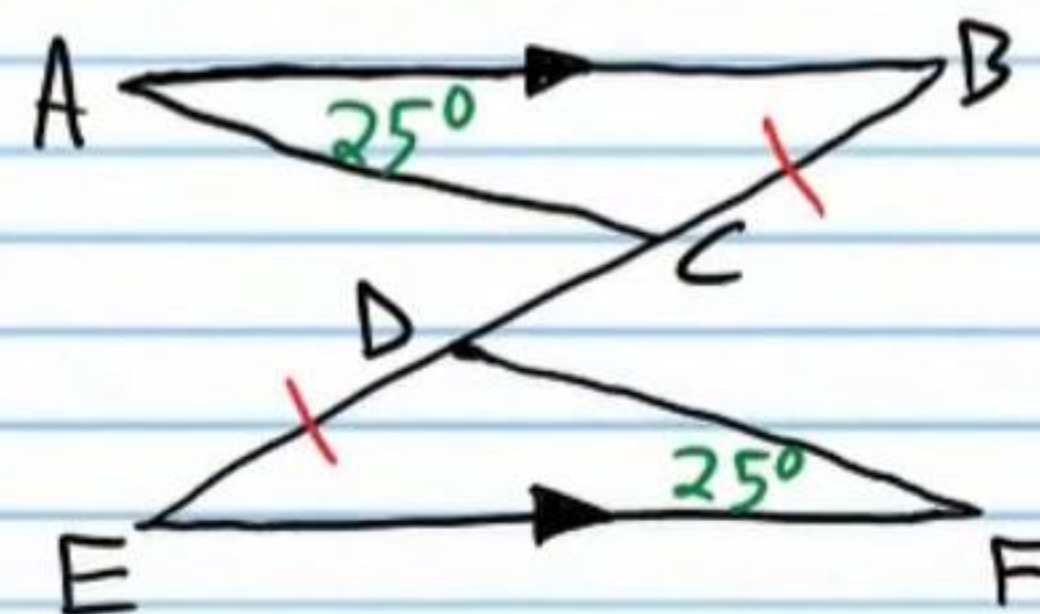
SAS

Show that each statement below is true by writing a two-column proof.

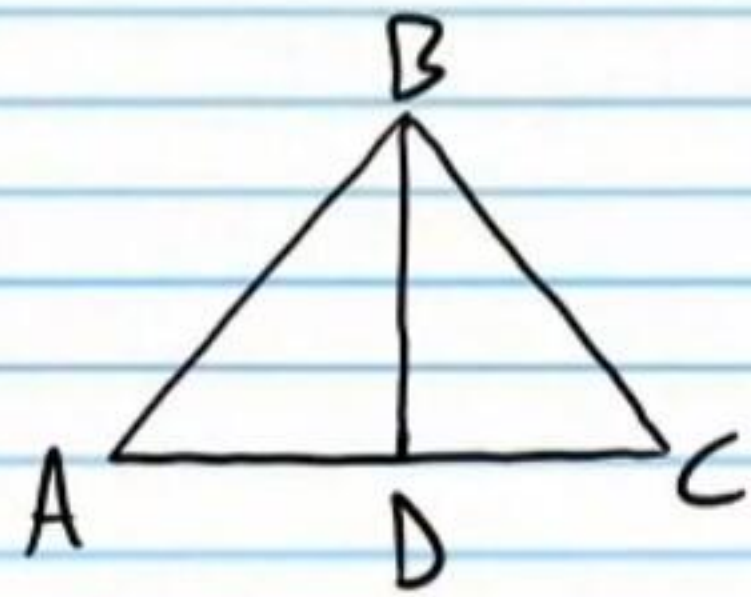
② If C is the midpoint of $\overline{AD}$ and $\overline{BE}$, then $\triangle ABC \cong \triangle DEC$.



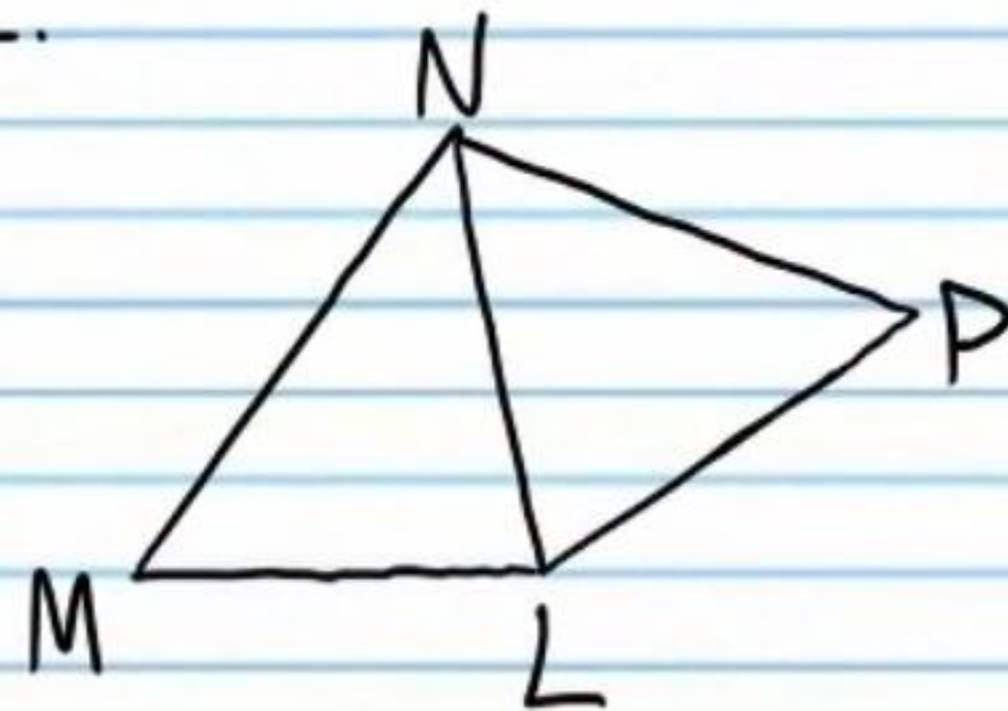
② If $\overline{AB} \parallel \overline{EF}$ and $\overline{DE} \cong \overline{CB}$, then $\triangle ABC \cong \triangle FED$.



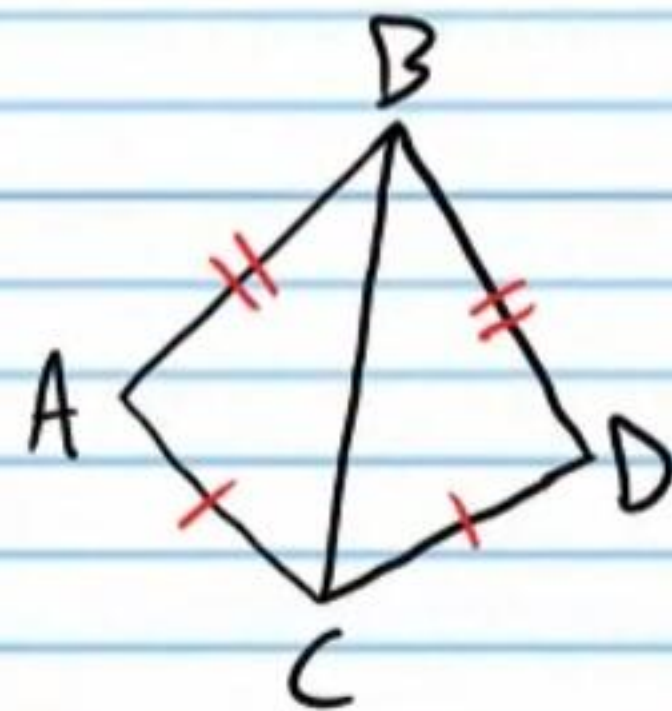
②⑦ If $\overline{AC} \perp \overline{BD}$ and $\angle ABD \cong \angle CBD$, then $\triangle ABD \cong \triangle CBD$.



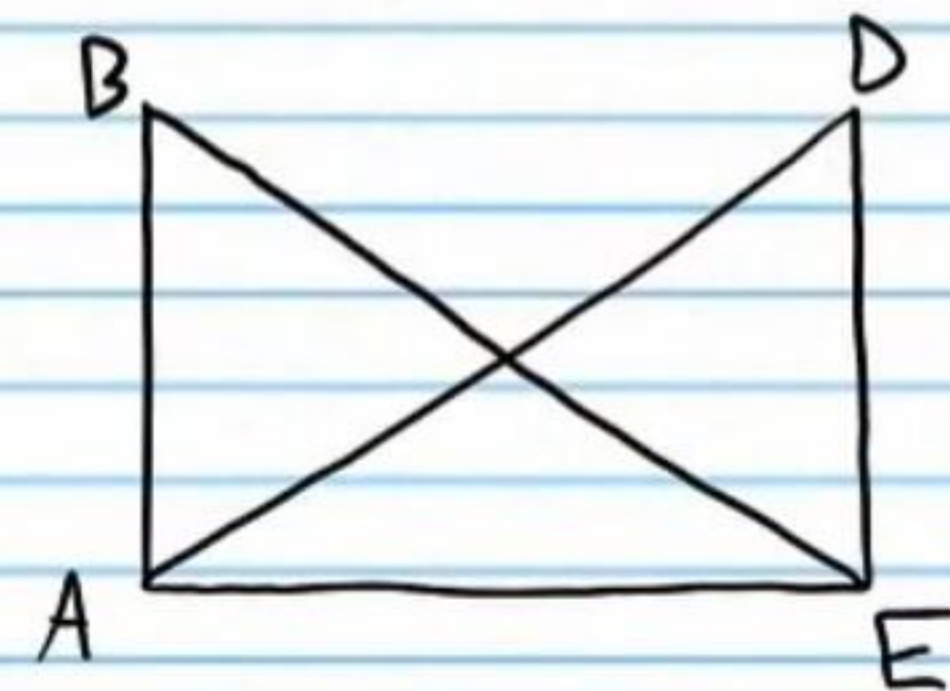
②⑧ If $\overline{NL}$ bisects $\angle MLP$ and $\overline{ML} \cong \overline{LP}$, then $\triangle MNL \cong \triangle PNL$.



②⑨ If $\overline{AB} \cong \overline{DB}$ and $\overline{AC} \cong \overline{DC}$, then $\triangle ABC \cong \triangle DBC$.



③⑩ If $\overline{AB} \perp \overline{AE}$ and $\overline{DE} \perp \overline{AE}$ and $\overline{AB} \cong \overline{DE}$, then $\triangle ABE \cong \triangle EDA$.



Answers for Proofs (Homework)

(25)	Statement	Reason
I)	C is the midpoint of $\overline{BE}$ and $\overline{AD}$.	Given
II)	$\overline{BC} \cong \overline{CE}$ and $\overline{AC} \cong \overline{CD}$.	Def. of midpoint.
III)	$\angle BCA \cong \angle ECD$	Vert. $\angle$'s Thm.
IV)	$\triangle ABC \cong \triangle DEC$	SAS

(26)	Statement	Reason
I)	$\overline{AB} \parallel \overline{EF}$	Given.
II)	$\angle ABC \cong \angle FED$.	Alt. Int. $\angle$'s Thm.
III)	$m\angle BAC = 25^\circ$ and $m\angle EFD = 25^\circ$	Given (diagram).
IV)	$m\angle BAC = m\angle EFD$	Trans. Prop of =.
V)	$\angle BAC \cong \angle EFD$	Def. of $\cong \angle$'s.
VI)	$\overline{DE} \cong \overline{CB}$	Given.
VII)	$\triangle ABC \cong \triangle FED$.	AAS

(27)	Statement	Reason
I)	$\overline{AC} \perp \overline{BD}$.	Given.
II)	$\angle ADB$ and $\angle CDB$ are rt $\angle$'s.	Def. of $\perp$.
III)	$\angle ADB \cong \angle CDB$	Right $\angle \cong$ Thm.
IV)	$\angle ABD \cong \angle CBD$	Given.
V)	$\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$.
VI)	$\triangle ABD \cong \triangle CBD$.	ASA

(28)	Statement	Reason
I)	$\overline{NL}$ bisects $\angle MLP$.	Given.
II)	$\angle MLN \cong \angle PLN$.	Def. of $\angle$ bisector.
III)	$\overline{NL} \cong \overline{NL}$	Ref. Prop of $\cong$.
IV)	$\overline{ML} \cong \overline{LP}$	Given.
V)	$\triangle MNL \cong \triangle PNL$	SAS

(29)	Statement	Reason
I)	$\overline{AB} \cong \overline{DB}$ and $\overline{AC} \cong \overline{DC}$.	Given.
II)	$\overline{BC} \cong \overline{BC}$.	Ref. Prop. of $\cong$.
III)	$\triangle ABC \cong \triangle DBC$.	SSS

(30)	Statement	Reason
I)	$\overline{AB} \perp \overline{AE}$ and $\overline{DE} \perp \overline{AE}$.	Given.
II)	$\overline{AE} \cong \overline{AE}$.	Ref. Prop. of $\cong$.
III)	$\overline{AB} \cong \overline{DE}$	Given.
IV)	$\triangle ABE \cong \triangle EDA$.	SAS